

Mission 005 Architect Handoff

Updated architecture assessment

****Verdict: acceptable with caveats****

The implementation now matches the fixed Mission 005 direction in the important places: Neon-backed persistence exists, agents write through a simple backend endpoint, reads go through DB-backed APIs when the DB is configured, and the UI remains read-only with freshness/lag visibility preserved.

What matches the fixed architecture

- Neon is the canonical production/preview database target.
- `POST /api/agent-events` is present as the simple agent write path.
- The write path is intentionally simple for v1: server-side payload validation plus canonical DB writes, with no write secret or signing layer.
- `GET /api/dashboard` and `GET /api/missions/latest` preserve their response envelopes while reading from the DB-backed store.
- Freshness states remain visible: `fresh`, `partial`, `delayed`, `stale`, `empty`.
- The app remains read-only for end users.

Key caveats / gaps

1. ****JSON fallback still exists when DB env vars are missing.****
 - That is fine for local/dev, but it must not remain the canonical path in production/preview after cutover.
2. ****Migration/backfill boundary is not fully closed yet.****
 - The schema exists, but the one-time JSON-to-Neon backfill and post-cutover no-JSON-truth guarantee still need operational confirmation.
3. ****Freshness is correctly surfaced at read time, but the DB also stores a freshness field on events.****
 - Treat that as diagnostic/denormalized metadata, not a second source of truth.
4. ****Backend handoff is still missing from the Mission 005 handoff set.****
 - That should be created as soon as schema/write/read implementation is finalized.

Canonical trust boundary

- ****Trusted:**** Neon rows, backend validation, read API shaping.
- ****Untrusted:**** client state, agent payloads before server validation, JSON files/feeds except as seed/export artifacts.
- The canonical system of record is Neon + backend validation, not JSON replication.

Freshness semantics

- `updatedAt` and `lagMs` should remain DB-derived at read time.
- Keep the states visible and meaningful: `fresh`, `partial`, `delayed`, `stale`, `empty`.
- UI should continue to show updated-at and lag explicitly.

Migration / backfill boundary

- JSON may be used once for bootstrap/backfill.
- After cutover, all new writes must go only to the simple backend API and Neon.
- Any JSON artifact must be produced from DB state, not vice versa.
- Production/preview should not depend on JSON fallback once Neon is live.

Coordinator handoff

Proceed with Backend, Frontend, and QA execution, but treat final cutover as pending until backend confirms the DB-backed runtime path and QA verifies the live environment is not using JSON as the truth source.